Appendix 1

Population Pharmacokinetic Analysis Guidance

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ABBREVIATIONS AND DEFINITIONS OF TERMS

Abbreviation	Definition
AIC	Akaike Information Criteria
ADDL	Additional dose flag
AUC	Area under the curve
BLQ	Below Limit of Quantification
BMI	Body Mass Index
CI	Confidence Intervals
CL	Clearance
Cmax	Maximal concentration
CRF	Case Record Form
CTS	Clinical Trial Simulation
CWRES	Conditional Weighted Residuals
CWRESI	Conditional Weighted Residuals with Interaction
DES	Differential Equations System
df	Degrees of Freedom
DRF	Data Request Form
DV	Dependent Variable
EBE	Empirical Bayes Estimates
EM	Expectation maximization
EPS (ε)	Residual variability
EST	Estimation ($ESTIMATION)
ETA (η)	Empirical Bayes prediction of the interindividual random effect
EVID	Event Identification variable
FME	Full Model Estimation
FO	First-Order Estimation Method
FOCE	First-Order Conditional Estimation Method
FOCEI	First-Order Conditional Estimation with Interaction Method
GAM	Generalized Additive Model
II	Interdose interval
IIV	Interindividual Variance
IOV	Interoccasional Variance
IPRED	Individual Predicted value based on individual's ETAs
IWRES	Individual Weighted Residuals
Ka	Absorption rate constant
Lasso	Least absolute shrinkage and selection operator
LRT	Likelihood Ratio Test
MDV	Missing Dependent Variable
MLE	Maximum Likelihood Estimation
NONMEM	Nonlinear Mixed Effects Modeling
NPDE	Normalized Prediction Distribution Error
NSIG	Number of significant digits
NPC	Numerical Predictive Check
OBS	Observations
OFV	Objective Function Value
OMEGA (ω^2)	Intersubject Variance

Abbreviation	Definition
PBPK	Physiological Based Pharmacokinetic
PD	Pharmacodynamic
PK	Pharmacokinetic
Pop PK	Population Pharmacokinetic
PMAP	Population Modeling Analysis Plan
PMAR	Population Modeling Analysis Report
PPC	Posterior predictive check
PRED	Population Predicted Value
PREDPP	Predictions for Population Pharmacokinetics
PsN	Perl Speaks NONMEM
Q	Intercompartmental Clearance (central-peripheral)
SIGMA σ^2	Residual Variance
SCM	Stepwise Covariate Modeling
SS	Steady-state flag
TAD	Time After Dose
THETA (θ)	Estimate of Fixed Effect
USPI	United States Prescribing Information
V	Volume of Distribution
VPC	Visual Predictive Check
WAM	Wald's approximation to the Likelihood Ratio Test
WRES	Weighted Residuals

1. Purpose

The purpose of this chapter is to provide guidance for conducting population pharmacokinetic (Pop PK) analyses in order to streamline and optimize practices across the clinical drug development spectrum. All Pfizer colleagues and external service providers who plan to conduct Pop PK analyses are encouraged to read this chapter in combination with the relevant regulatory guidances and the Pfizer internal document: the Points for Consideration: Statistical Aspects of Population Pharmacokinetic and Pharmacodynamic Modeling [1], which include specific strategic and technical advice [2, 3].

2. Scope

The scope and focus of this guidance is to provide recommendations for conducting Pop PK analyses, however, many of the principles outlined within this chapter will also be applicable to pharmacokinetic/pharmacodynamic (PK/PD) analyses. The reader is reminded that population modeling is an evolving and interdisciplinary science which encompasses statistical, mathematical, pharmacological, and clinical principles. It follows then that these recommendations are based on the current state of knowledge, and flexibility should be granted as methodological developments and enhancements to the population modeling process evolve. Providing in-depth technical or theoretical guidance, therefore, is beyond the scope of this chapter, and readers are strongly encouraged to seek technical advice from references that are listed under each section in this guidance document.

3. Background

Population pharmacokinetics is defined as the study of the variability in drug concentrations among individuals, including healthy volunteers and target patients [2, 3]. Pop PK analyses are an integral part of model-based drug development [4] and have become a regular component of regulatory submissions. Pop PK analyses are performed to characterize the PK of a compound and to estimate the impact of physiological and demographic factors that may affect the PK. The results of Pop PK analyses are often included in the drug label typically supporting the dosing recommendations in special populations (e.g., renally-impaired, elderly, etc.). A population analysis is most frequently performed using nonlinear mixed effect modeling implemented in NONMEM (ICON Development Solutions, Ellicott City, MD) [5]. Much of the information contained within this guidance will refer specifically to analyses conducted with NONMEM.

Because there has been a dramatic increase in both the quantity and quality of Pop PK analyses, the pharmaceutical industry has increasingly recognized the importance and necessity of implementing a systematic approach to these analyses [1]. Considering that the nature of population modeling is an evolving and interdisciplinary science, it is unrealistic to expect that a consensus will be reached on every aspect of the analysis. However, the importance of improved efficiency through streamlined and standardized approaches should not be underestimated, as population modeling is an iterative process through “learning” and “confirming” phases of clinical drug development involving years of time and multiple analysts [6].

The recommendations are described in the following sections: 4. Points to Consider before Conducting a Population Pharmacokinetic Analysis; 5. Population Pharmacokinetic Base Model; 6. Population Pharmacokinetic Final Model; 7. Graphical and Numerical Diagnostics.

4. Points to Consider before Conducting a Population Pharmacokinetic Analysis

Before conducting a Pop PK analysis, it is required that a population modeling analysis plan (PMAP) be developed.[1] The analyst should strive for a proper balance in the level of specificity in the PMAP in order for it to be an effective planning tool. A PMAP that is too vague provides little information to guide model development, requiring the analyst to spend a fair amount of time writing up the methodology after completion of the analysis. On the other hand, a PMAP that is too specific may result in an actual analysis that bears little resemblance to the plan, and again, a fair amount of effort may be required after completion of the analysis to document the deviations from the analysis plan. In some cases, the PMAP can be considered a 'living document' and can be updated as more information becomes available. For Phase 2/3 analyses, however, the PMAP should be finalized prior to database lock.

During development of a PMAP, the choice of studies to be included in the Pop PK analysis depends on the purpose of the modeling exercise. Phase 1 and Phase 2a studies typically include intensive PK sampling and provide data that enable development of a robust structural model. Phase 2b/3 studies typically include sparse PK data and a more representative covariate distribution that would provide information important for covariate model building. A well-prepared PMAP should provide an overview of the purpose of the modeling exercise and the studies to be included in the analysis.

Once the choice of studies for the Pop PK analysis is made, careful thought must go into completion of the Data Request Form (DRF). The analyst should attempt to include any variables that may be of potential use in future analyses to ease pooling of datasets. For example, the analyst should consider the use of variables, such as ADDL, SS, or II in early development, as having them already included in these datasets will facilitate pooling of data and efficient delivery of the updated dataset at later times. Thus, consideration of future needs should be made at the time of the original DRF creation.

When draft data become available, it is recommended that thorough data checks be conducted in advance of analyses to eliminate potential errors. In addition, exploratory data analyses (graphical and/or statistical summary) should be performed to help determine the base structural model and the components of the statistical model, as well as, identify potential covariates relationships and outliers. Summary statistics of the dependent variable (DV) and demographics (including covariates) should be checked for any abnormal values. Graphical exploration of a PK dataset should include, but is not limited to, distribution and correlation of covariates, and observed concentration-time profiles with median values at nominal time points by key design features, such as study and treatment groups (e.g., dose/regimen groups).

Once the planned, definitive analysis is completed, the results should be summarized in a Population Modeling Analysis Report (PMAR). It is important to note that assumptions and decisions made during model development be clearly documented in the PMAP and PMAR.

5. Population Pharmacokinetic Base Model

5.1 Definition

The Pop PK base model is the structural form of the model, which may include structural covariates for the fixed effects. In addition, the base Pop PK model should include the specification of the interindividual and residual random effects, as well as, the corresponding covariance structures.

The development of a Pop PK base model includes two steps:

- Structural model development
- Statistical model development

5.2 Inferences from Population Pharmacokinetic Base Models

Since the Pop PK base model should adequately capture important characteristics of the concentration-time profiles of interest, it is adequate for making basic inferences about exposure levels and variability (interindividual, interoccasion, and residual). Once the Pop PK base model is defined, it is then possible to proceed to further covariate model building (in addition to any structural covariates which have been included in the base model), which examines demographic, pathophysiological, environmental, or concomitant drug-related factors that could further explain sources of interindividual variance (IIV).

5.3 Data

A few recommendations are provided as guidance for handling data-related issues during the base model development in the following sections, however, it is understood that each modeling exercise may require special consideration and, hence, the analyst should treat the data accordingly.

5.3.1 Full versus Validation Datasets

The analyst should consider including as much information as possible in the full dataset during the model building exercise in order to obtain the most robust model. Model evaluation examines the predictability of the developed model, and it is recommended that an external validation dataset or a bootstrapping procedure be used for this purpose. An external validation dataset is one that was not available at the time of model building and parameter estimation. Splitting of the full dataset used in the modeling exercise in order to build a validation dataset is not recommended.

5.3.2 Below Limit of Quantification Concentration Data

When data are below the limit of quantification (BLQ) of the bioanalytical assay, the analyst must understand that the method chosen for handling these samples may influence the estimation of PK parameters. Common approaches for handling concentrations that are BLQ have been shown to introduce bias in the parameter estimates [7-9]. A comprehensive overview outlining approaches for fitting Pop PK models in the presence of BLQ data [10] was published by Stuart Beal.

The analyst should first determine what percentage of the entire dataset is BLQ. In general, ≤10% of samples which are BLQ may be considered acceptable, however, the analyst should check the distribution of data to ensure that there are no systematic trends present [7]. In certain

instances, it may be necessary to reconsider the acceptable limit of 10% of the data as BLQ, and this should be considered on a case-by-case basis.

If the proportion of samples which are BLQ in the entire dataset is less than 10%, and there are no systematic trends observed, then ignoring these samples will typically not have a significant impact on the structural model development. Therefore, when BLQ records are present in the dataset, the MDV and EVID records should be set to 1 and 2, respectively, so that estimated values can be computed for BLQ records.

When a substantial proportion of BLQ data are present (≥10%), treating these samples as missing may lead to biased PK parameters, such as underestimated clearance (CL) and overestimated volume of distribution (V) values [7, 8, 10-12]. In these instances, the recommended approach is based on simultaneous modeling of continuous and categorical data, in which the BLQ observations are treated as categorical data [10-13]. This method is commonly referred to as the M3 approach [10]. For the M3 approach, both the MDV and EVID records should be set to 0 for the BLQ records.

5.3.3 Missing Concentration, Sample, and Dosing Time Data

Missing data may include the following; 1) missing concentration data, 2) missing PK sample times, 3) missing dosing information (dose administration time or actual dose administered), or 4) missing covariate information. This section will only discuss the first three; missing covariate information will be discussed in 6.4.2 Missing Covariates.

In well-controlled Phase 1 or Phase 2a studies, missing data are not as commonly observed as in larger, late-stage, Phase 2b or Phase 3 studies. If data are missing from Phase 1 or Phase 2a studies, discussions with the study team in an attempt to retrieve this information is highly encouraged. However, this may not be feasible in the larger, late-stage patient studies and, therefore, ways to handle this missing data are discussed here. It is vital that the analyst be involved in all aspects of protocol development and study conduct in order to ensure the integrity of the data.

A missing PK concentration should be dropped from the dataset unless it is a BLQ record described above. When the time of sample collection is missing or inappropriately recorded, problems may be encountered when attempting to relate the sample time to the time of drug administration. Data checks should be in place during the conduct of a study to identify these erroneous collection times and should include the following: ensure that the PK collection times are recorded in 24-hour clock time; ensure that PK collection dates and times are recorded in sequential order; and confirm/query all blank data fields in Case Report Forms (CRF). The analyst is encouraged to work with the study team, data managers, and programmers to resolve these issues, preferably before the database is locked. Missing dosing data (actual time of dose administration or the actual dose amount given) can be the result of inaccurate source documentation at the clinical site. As mentioned above, it is strongly encouraged that these data be retrieved.

If the data cannot be retrieved, the analyst will need to consider a few options to handle these missing data. One consideration would be to use nominal PK collection times or the planned

administered dose in the absence of actual collection times or dosing information, respectively. However, the analyst is encouraged to carefully consider this option, as this approach could result in biased parameter estimates. Other options include dropping the data from the dataset. In any case, it is recommended that procedures for handling the missing data be documented in the PMAR, sensitivity tests be performed to test any assumptions made, and the impact on parameter estimates be evaluated and reported.

5.3.4 Ordinary versus Log Domain

It is recommended that the data be log-transformed and that an additive error model be used, at least initially, in the log space. Log-transformation may add stability to the model and parameter estimates, particularly when there is a wide range of concentrations; i.e., hundred- to thousand-fold. The analyst can consider modeling in the ordinary domain; however, this should be addressed on a case-by-case basis.

5.4 Assessment of Model Adequacy

At all stages of the model development (base, full covariate and final), models should be evaluated using the following criteria:

- **Successful Minimization:** The analyst must develop a model capable of achieving a successful minimization status,
- **Visual inspection of diagnostic plots:** See Section 7. Graphical and Numerical Diagnostics
- **Likelihood Ratio Test (LRT):** While the LRT is a numerical way of evaluating nested models, using its value as a sole criterion for structural base model selection is not recommended due to an inflated Type I error [13, 14].

Good PK modeling practice dictates that analysts always fit all models with multiple sets (3 at a minimum) of perturbed initial estimates to minimize the likelihood of being caught in a local minimum.

The question of what magnitude of change in Objective Function Value (OFV) should be used for assessing addition of parameters in competing models often arises. Maximum likelihood theory states that the difference in OFV between competing nested models is χ^2-distributed, with degrees of freedom (df) equal to the difference in the number of estimated parameters between the competing models. It has been shown that this assumption can be estimation method-dependent (at least in NONMEM) and often does not retain the nominal Type I error rate [14]. For this reason, the analyst should refrain from assigning statistical significance; i.e., a p-value, to describe differences between competing nested models.

While it is recommended that the LRT not be used for assigning statistical significance to model differences, this does not imply that the magnitude of OFV differences is not informative. As a general rule, OFV changes of greater than 10 for one additional parameter are likely to be associated with an improved model fit. If there is a strong biologic/physiologic rationale for inclusion of a particular parameter, the criterion could be relaxed to a smaller value.

For non-nested models the analyst can use an analogue of the OFV, the Akaike Information Criteria (AIC), to evaluate differences in competing models. This is calculated as the OFV + 2*p, where p is the number of estimated parameters in the model. The model with the lower AIC is considered to be superior.

5.5 Structural Model Development

The first step in Pop PK model development is to focus on the structural form of the base model, such as 1-compartment versus 2-compartment PK models. The analyst should use any prior knowledge to assist in the development of the structural form of the model. The choice of the model should be data driven and depends upon the purpose and intended use of the model, as well as the questions that the modeling exercise is trying to answer. While the change in the OFV relative to the change in number of parameters is an important way of evaluating nested models, using this value as a single guide for structural base model selection is not recommended due to inflated Type I error as discussed above [13, 14].

5.5.1 Structural versus Error Model

It is recommended that the analyst first refine the base structural model and then subsequently, refine the residual error model. When refining the structural base model, the analyst should postulate a reasonable residual error model. Since it is recommended that all data be log-transformed, an additive error model in the log space is recommended during this step in base model development. Upon inspection of residual diagnostic plots, if the assumptions of the postulated residual error model appear to be violated, it is appropriate at this time to refine the residual error model as discussed further in Section 5.6.4 Residual Error Model.

5.5.2 Parameterization

It is recommended that the PK parameters, CL and V, be included as the default parameterization using PREDPP subroutines. This will allow for ease of explanation of parameter estimates and covariate model building, as well as reduced computation time. In some situations, differential equations may need to be used for parameterization using $DES.

5.5.3 Structural Model Covariates

It is recommended that structural covariate parameters be included as part of the base structural model development to ensure model stability when such covariates are known a priori to be highly influential. For example, it may be appropriate to treat creatinine clearance as a structural model covariate for CL in the base PK model for drugs that are primarily cleared by the kidneys. Subsequent covariate model building would then start from the base model which includes an effect of creatinine clearance on CL. If the clearance of a drug is known to be impacted by genotype, that information should also be included as a covariate in the base model. Parameters which are highly influential on the absorption of the drug, including food or formulation effects, if not accounted for, may lead to model instability. In addition, an understanding of the disposition of the drug can also aid in structural model development. For example, if a drug is partially eliminated by the kidneys, the analyst can empirically consider expressing the CL as renal and non-renal clearance components in the base structural model, with an effect of creatinine clearance on renal CL. This will allow for a mechanistic understanding of the elimination pathway of the drug.

Allometric scaling can be considered a component of the PK base structural model, particularly when there is a wide range of body weights, or the inclusion of both adult and pediatric data. Initially, the analyst could consider fixing the values (as a power function) to 0.75 for CL and 1.0 for V, however, this should not preclude the analyst from estimating these values if this approach results in a better model fit. It is recognized that there are other methods to handle scaling of PK parameters for pediatric subjects, including maturation models and Physiologically Based Pharmacokinetic (PBPK) models.

5.6 Random Effects Model Development

5.6.1 General Principles

It is recommended that the interindividual variance, Ω, with elements ω^2 for each of the associated PK parameters, be explored and refined through the modeling effort. Since the primary tool used for Pop PK modeling is NONMEM, by convention, ETA will represent these elements. Likewise, residual variance sigma, σ^2, with elements ε, should be introduced into the population model to partition random error to interindividual and residual components. Again, since modeling is usually performed using NONMEM, by convention, EPS will represent residual variance elements.

Modeling residual variance without introducing interindividual variance on PK parameters is equivalent to a naïve pooled approach, in which both the interindividual and residual variances are estimated combined into one variance. Following the initial successful introduction of ETA on the selected parameters, different residual error models can be further investigated. Once a reasonable residual error model has been defined, both the IIV and residual error models should be further refined as an iterative process so that model performance can be improved.

5.6.2 Interindividual Variability

In general, it is recommended that IIV be implemented using an exponential function in order to maintain positive PK parameters. The IIV can be introduced on the PK parameters for which the estimation of variability can be supported by the data. For example, CL and V are the PK parameters for which IIV is most commonly introduced. If there are rich data in the absorption phase of the concentration-time profile, ETA may be placed on the parameter describing the absorption process, such as the absorption rate constant (Ka). Unless PK data are available from both intravenous and oral administration in the same subjects, it may be difficult to characterize the variability for oral bioavailability.

It is preferable to parameterize V in terms of central and peripheral volumes rather than volume of distribution at steady-state for a 2-compartment PK model, because covariate effects may be more identifiable and interpretable. Sharing one ETA for central and peripheral volumes may be implemented, but the assumption should be reasonable for the compound.

Use of the OFV as the only criteria for inclusion/exclusion of ETA is not recommended. The LRT for a non-zero variance component is a non-standard one, in which the null hypothesis places the true value of the variance parameter (0) on the boundary of the parameter space defined by the alternative hypothesis [15]. Under such circumstances, the χ^2 random variable with the number of degrees of freedom (df) for the test equal to the increase in the number of variance and covariance terms is asymptotically conservative and leads to an inflated Type 1

error. The bias could be substantial when information is not sufficient and many random effect terms are added simultaneously.

It should be recognized that an extremely small estimate of IIV approaching zero does not necessarily indicate that there is a lack of IIV for the parameter, rather it may simply suggest that the data are not robust enough to estimate the IIV or there is shrinkage in the variance due to the sparse data [16]. In these circumstances, it is recommended that the IIV be fixed to zero when using maximum likelihood methods. However, the Expectation Maximization (EM) estimation algorithms available in NONMEM 7 are inefficient when IIVs are fixed to 0. An alternative in this case is to fix the IIV to a small value to maintain efficiency of these algorithms. Further details on the impact of shrinkage on diagnostics are discussed in 7.7 Empirical Bayes Estimates-Based Diagnostics.

5.6.3 Off-Diagonal Elements of OMEGA Matrix

It is recommended that introduction of an omega block be delayed until structural covariate parameters and/or allometric scaling have been considered. Once structural covariates are included in the base model, it is recommended that consideration be given to fitting a full block omega-structure on the base model, followed by inspection of the correlations among the ETAs to guide the development of a parsimonious omega-structure. Characterization of the full omega block structure may be limited by the informative nature of the data; however, at a minimum, it is recommended that the correlation between CL and V be examined. Assuming a diagonal omega-structure when correlations exist among the interindividual random effects, may lead to unrealistic combinations of parameters within or across individuals. It is also important to revisit the omega block structure after the covariate model building step, because a correlation that existed between ETAs before the covariate model building step may be explained by the addition of a covariate.

5.6.4 Residual Error Model

It is recommended that the concentration data be log-transformed to stabilize the residual variability. When working in the log domain, an additive error model should be used. An example of implementation of a combined residual error (additive and proportional) model in the log domain can also be found in the following reference [17].

Implementation of the "Thetarized SIGMA" is currently the only way to calculate Individual Weighted Residual (IWRES) values. Current versions of NONMEM do not store the IWRES variable; therefore, these values must be generated by incorporating explicit terminology into the control stream. As this is a critical vector for assessing residual error performance, it is recommended that thetarized SIGMA be used as the default until a future version of NONMEM allows for direct calculation of these values [1].

5.6.5 Interoccasion Variability

The importance and need for an interoccasion variability (IOV) component in the model depends primarily on the study design; i.e., crossover trials [18, 19]. IOV may be introduced before covariate model building; i.e., in the base model; however, it is recommended that the characteristics of IOV be estimated throughout the modeling process. Occasion should be defined by study design features and should be data driven. In Phase 1 crossover, dose-

escalation studies, occasion is often defined by period. In long-term late phase studies, occasion may be defined as the time frame over which PK parameters could change; e.g., weeks, months, years, etc. Careful consideration should be given when pooling data from different phases, as this approach could result in inconsistent definition of occasions across studies, making interpretation of IOV difficult. It should be noted that inclusion of IOV into the Pop PK model may be limited in terms of future predictive capacity.

5.6.6 Covariance Step

One of the key outputs from the covariance step ($COV) is the covariance matrix of parameter estimates. The accuracy of this covariance estimate increases as the number of (statistically independent) observations increases. The covariance estimate provides certain important quantitative information about the precision of the parameter estimates; however, it is widely agreed that this type of information can better be obtained by a bootstrapping method, which alleviates the concern about the potential asymmetrical distribution of parameters or non-asymptotic assumptions [20-22]. A bootstrapping approach allows the empirical distribution of parameter estimates to be readily obtained from a large number of successful runs (either successful minimization or termination due to rounding errors). To provide a robust representation of the parameter distribution from the bootstrap procedure, it is recommended that appropriate stratification be considered on relevant study design characteristics and covariates; e.g., dose, gender, patients versus healthy volunteers, etc.

When the $COV step runs successfully, further inspection of the correlation matrix of the estimates is recommended. Oftentimes, when one or more correlations are near 1, the model may still be over-parameterized and unstable (the $COV step "successful" runs may be because a local minimum is reached).

When the $COV step fails, it often implies that the model is over-parameterized, therefore, the analyst should strive to build models with a successful $COV step. If the $COV step fails, simplification of the structural and/or statistical aspects of the model may be necessary. In some cases, trying different initial estimates may help the $COV step to converge; however, this is often indicative of reaching a local minimum. It should also be recognized that successful $COV runs may not always be possible due to limitations in the data. Under such circumstances, a successful run with failed $COV step should not be discounted because useful information about parameter estimates is still provided.

5.6.7 Eigenvalues of the Correlation Matrix

Another key output from the $COV (with option PRINT=E) are the Eigenvalues of the correlation matrix. The ratio of the maximum to the minimum Eigenvalue yields the condition number, which provides information about model stability. When a model generates a condition number that is greater than 1000, this indicates an ill-conditioned model often due to over-parameterization [23]. Inspection of the correlation matrix can provide insight into the source of the ill-conditioning. It is important to note that VPC diagnostic plots will not identify over-parameterization. Rather, the more parameters a model allows, the better the VPC will appear. It is understood that a model with a condition number larger than 1000 may have applications in certain scenarios, such as a modeling exercise when the main objective is to quantify uncertainty or to build a mechanistic model despite limited information.

6. Population Pharmacokinetic Final Model

6.1 Definition

The final Pop PK model is the most parsimonious model for which all relevant covariates of interest have been evaluated and retained. This section will describe two approaches that are recommended for inclusion and evaluation of covariates in order to arrive at the final model.

The full covariate model is defined herein as that model which contains all relevant covariates of interest irrespective of the approach that was implemented to evaluate the model. This definition can be somewhat at odds with that provided in the statistical literature [24], which defines the full model as that model for which all pre-specified covariate parameters are included (estimated) simultaneously, an approach described herein as one of the recommended covariate model building *procedures* that may be used to arrive at the final model (see Section 6.3 General Principles). For clarity, note that this approach is defined as the full model estimation (FME) procedure in this guidance [24].

6.2 Inferences from the Population Pharmacokinetic Final Model

The inferences drawn from the final Pop PK model include primarily the quantitative assessments of how precisely covariate effects can be estimated in the model, and ideally, an assessment of the clinical relevance of those effects.

6.3 General Principles

The final Pop PK model should adequately describe the PK of a compound and it should seek to describe the sources of variability in the PK. Variability in PK may be described by interindividual, interoccasion, and residual effects, which may be identified and characterized via addition of covariates to the model. Two primary approaches for inclusion of covariates into the Pop PK model are described herein.

The first approach utilizes simultaneous inclusion of all covariates of interest into the model, thus, generating the final model. Using this approach, the analysis can be conducted quickly as it allows for estimation of all covariate effects at once. This approach includes the FME procedure described below.

The second approach utilizes an intermediate step, evaluation of a full covariate model, prior to arriving at a final model. When utilizing covariate evaluation procedures that implement full covariate model approaches, it should be remembered that the final model is considered to be parsimonious to the extent that the minimum OFV is closer to the full covariate model OFV than the base model OFV, while containing as few covariate parameters as possible. Note that if the change in OFV (ΔOFV) between the base and full covariate model is very small relative to the number of covariate effects (i.e., in comparison to a χ^2-square distribution with df equal to the number of covariate effects in the full covariate model), it may not be necessary to build a full covariate model, therefore, the analyst might conclude that the base model is parsimonious. Examples of this approach include the SCM (stepwise covariate modeling) or WAM (Wald's approximation to the LRT) procedures described below.

Implementation of full covariate model approaches may provide benefits, such as useful diagnostic information contained within the covariance matrix of the estimates and the ability to assess outliers associated with extreme values of the covariates. In addition, full covariate model approaches may be useful for making inferences for covariate effects of interest that are not likely to be selected for inclusion in a parsimonious final model (e.g., ruling out drug-drug interaction effects).

On rare occasions, it may not be necessary to implement covariate model building procedures in order to obtain a parsimonious final model; however, these procedures are appropriate when the goal is to obtain a predictive model. A parsimonious final model which does not contain covariates may be more predictive than a model which does incorporate covariate effects, since it will smooth out noise due to "nuisance" covariate effects that may have no predictive value.

6.3.1 Covariate Model Building Procedures

The reason for including and evaluating covariates in the Pop PK model is to determine which covariates are responsible for the greatest reduction in the OFV. It is recommended that systematic procedures be incorporated for covariate model building to improve consistency and harmonization across analyses. It is important to be aware that each systematic covariate selection procedure has limitations. Although these selection procedures can result in models that fit well, there are no guarantees that the most parsimonious (or plausible) model will be obtained, especially if collinearity exists among the covariates and covariate parameters. It is important to ensure that the base model developed from the previous step is as stable as possible prior to covariate model building, as this may help to mitigate the problems that collinearity can have on covariate selection procedures.

The FME procedure is recommended for covariate model building to arrive at the final Pop PK model [24, 25]. This approach emphasizes parameter estimation (as opposed to stepwise hypothesis testing) and measures of estimation precision (% Relative Standard Error, bootstrap confidence intervals (CI)), allowing for the direct assessment of the clinical relevance of covariate effects; e.g., true lack of an effect versus lack of information about that effect. The FME procedure enables a direct assessment of all covariate relations of interest without the need to rely on data-driven model selection criteria. The FME procedure can also be conducted quickly as only a single run is needed, and the output is easy to interpret. On the other hand, correct inferences drawn from the output using the FME procedure assumes that the covariate relationships have been captured correctly and that no important covariates have been omitted.

The SCM procedure is commonly used for covariate model building, however, requires evaluation of a full covariate model prior to arriving at the final model. The SCM procedure includes a forward selection (resulting in a full covariate model) and a backward elimination process (resulting in a final model). The forward selection process within SCM is particularly prone to the effects of collinearity when two or more covariates (or covariate parameters) are highly influential, but only when they are included simultaneously in the model. Since the forward selection procedure *sequentially* evaluates the addition of one covariate effect at a time, it is possible that a particular combination of highly influential covariate effects may never be evaluated.

Additional covariate model building procedures exist, however, these are not generally recommended. The WAM is another procedure in which a full covariate model with successful generation of a covariance matrix must be developed prior to arriving at the final model [26]. The GAM (generalized additive modeling) procedure may not be appropriate for handling time-varying covariates or randomly time-varying parameters (i.e., not possible for assessing covariates on parameters that do not have interindividual variability) [27]. Finally, the Lasso (least absolute shrinkage and selection operator) procedure has occasionally been implemented [28]. A list of the most commonly used covariate model building procedures, along with the pros and cons for each, are outlined in Table 1 below.

Table 1. Covariate Model Building Procedures

Procedure	Definition	Pros	Cons
FME[24] (full model estimation with all inclusive covariates)	Estimate all predefined covariates of interest simultaneously and make inferences.	• Single run without model building • Output easy to interpret • Final model attained in one step	• Careful pre-specified selection of covariates necessary
SCM (stepwise covariate modeling)	Forward selection of covariates, followed by backward elimination.	• Easy concept/ implementation (with PsN)	• Selection bias • Multiple comparisons without correction • Must generate full covariate model prior to final model
WAM[26] (Wald's approximation to the LRT)	Full covariate model fit to calculate the approximate LRT for all possible restricted models. Final model selected based on the maximum (actual) SBC derived from the NONMEM model fits for the 10–15 most probable models.	• Fewer runs as with large numbers of covariates	• Requires a full covariate model with a covariance matrix prior to final model
GAM[27] (generalized additive modeling)	Individual parameter estimates (empirical Bayes estimates) are regressed on the covariates using a generalized linear additive model.	• Easy concept/ implementation	• Shrinkage • Correlations between parameters • Difficulty with time varying covariates
Lasso[28] (least absolute shrinkage and selection operator)	All covariates must be standardized so that mean = 0 and standard deviation = 1	• Superior to SCM in obtaining a predictive covariate model on a small dataset	• Benefit is negligible for large datasets

6.4 Covariates

6.4.1 Types of Covariates

The key challenge to building covariate models is the problem of over-parameterization, which can lead to convergence difficulties and $COV step failures. When two or more covariates are highly correlated (or imbalanced among discrete covariates), the analyst will need to make difficult choices to select a reduced set of covariates that have the greatest scientific plausibility. For that reason, covariates should be selected judiciously to reduce collinearity among

continuous covariates and imbalance among discrete covariates. Peck and Berk provide further discussion related to the pitfalls of covariate selection procedures when collinearity is present [29, 30].

It is imperative that the scientific (i.e., pharmacologic, biologic, pathophysiologic, clinical, etc.) merits of each covariate and covariate parameter be assessed prior to inclusion in covariate model building procedures. As an extreme example, it does not make sense to include sex as a covariate into the model if only females were present in the study population. Likewise, since body weight, body mass index (BMI), lean body mass, and height are all highly correlated parameters describing body size, it serves little purpose to include all of these as covariates into the model. On the other hand, there is a rational scientific basis for including laboratory measures of renal function if a portion of the subject population has renal impairment, even if the drug is not primarily cleared by the kidneys, as renal impairment may induce changes in protein binding and hepatic metabolism [31, 32].

It is recommended that the analyst start with graphical exploratory data analysis, non-compartmental parameters, such as area under the curve (AUC), maximal concentration (Cmax), and normalized concentration-by-dose profiles to isolate and reveal patterns for selecting pre-specified covariates for covariate model development. This should provide some additional confidence that there exists adequate rationale for including the pre-specified covariates into the modeling procedure.

There are a variety of covariates that may be considered for inclusion into the Pop PK model. A priori literature, preclinical, or early clinical evidence may exist to suggest that a covariate should be included on a structural model parameter in order to adequately describe the drug concentration profile, especially if a previous study showed a statistically significant effect of that covariate on drug exposure. For example, covariates known to impact CL, such as renal CL for drugs eliminated primarily by the kidneys, should be included into the base structural model. This topic has been previously discussed in Section 5.5.3 Structural Model Covariates.

The covariates selected for inclusion into the Pop PK model should be those covariates that have been identified as critical to test for impact on the drug concentration profiles based on a priori expectations as described above. These types of covariates may describe demographic, study design, laboratory, and/or other study/subject characteristics. For example, demographic covariates may include age, sex, race, ethnicity, measures of body size (weight, height, BMI, lean body mass, etc.), renal function, hepatic function, and genotype, among others. The study population, drug formulation, and whether the drug was administered with or without food are some study design characteristics that may need to be considered depending upon the circumstances. Laboratory values, such as creatinine clearance, may be important to include, as relevant, into the covariate model building procedure. Depending upon the drug and/or study population, information regarding genotype may be important to include as well. These examples are not intended to provide an all-inclusive, comprehensive list of covariates, and the analyst is highly encouraged to consider the totality of the scientific properties that are inherent to the compound and population under study when making decisions regarding covariate inclusion.

Finally, there may be occasion to examine covariates in an exploratory manner. The exploratory covariate analysis can be a useful tool for hypothesis generation; however, the results of these analyses should not generally be incorporated into PMARs which are intended for inclusion into regulatory submissions. There may be occasional exceptions, such as in the case of rare diseases or conditions (orphan diseases) which are manifest in only small numbers of subjects. In these instances, there may be only a single Phase 2b/3 trial with limited information available, necessitating a more exploratory approach to data analysis.

At this point in the Pop PK model building exercise, it is assumed that the correlation matrix of the parameter estimates from the $COV step output or bootstrapped output for the final base model was inspected to ensure that the structural model parameter estimates were not highly correlated prior to investigating the same covariates on multiple structural parameters. Typically, the effect of covariates is evaluated for CL and V unless there is some physiologically-based reason not to do so. Although the interpretability of most covariate effects on intercompartmental clearance and peripheral volume of distribution may be difficult, this should not preclude inclusion of covariates effects on these parameters if there is a sound scientific rationale to do so.

6.4.2 Missing Covariates

Every attempt should be made to retrieve covariate data that are missing from the dataset; however, it is recognized that there may be rare occasions when the data simply cannot be recovered. The percentage of missing data that is acceptable will depend on the nature of the covariate (continuous versus categorical, time-constant versus time-varying, etc.), how important that covariate is to the inference of the analysis, and other factors.

The manner in which missing covariate data are handled will most likely vary by project. Some potential options for handling missing covariate information are provided below:

1. Omit all records for the subject with the missing covariates.
2. Impute the missing covariates as follows:
    a. Set the covariate value to the study population mean or median (this approach may not have much impact if covariates are centered upon inclusion into the model). For example, the analyst could set missing body weight data to the mean or median of the study population, using sex-specific values if both males and females are present in the dataset,
    b. Set the covariate value to that subject's previous value (if available) with the assumption that the covariate value does not change over time,
    c. Set the covariate value to an interpolated value between the previous and subsequent values (if available).

Further details regarding single and multiple imputation procedures can be found in the Pfizer internal document: Points for Consideration document [1].

In any case, it is recommended that procedures for handling missing covariates be specified and documented in the PMAP and PMAR, respectively. Sensitivity analyses should also be

performed to evaluate the impact of the procedures for handling missing covariates on parameter estimates.

Time-varying covariates require special consideration. These types of covariates may not be routinely incorporated into the Pop PK model, but the analyst needs to be aware of the difference between including a baseline covariate on CL which is time-constant (e.g., sex, race) versus a covariate which has the potential to change over time. Time-varying covariates take on more importance with studies of long duration because covariates, such as body weight, certain laboratory parameters, or even age (e.g., extended pediatric studies) can change over the course of the study. In addition, the analyst is highly encouraged to carefully consider methods for coding time-varying covariates in the dataset (e.g., whether to carry the value forward, how far forward, etc.), as these issues can lead to a large amount of rework in data set preparation if not addressed prior to the analysis. Further details can be found in the paper by Wahlby [33].

6.4.3 Implementation of Covariate Models

Implementation of covariates into the Pop PK model should be done in as simple a manner as the data will allow and should describe the central tendency of the data. Generally the median (continuous) or most prevalent (categorical) covariate values are recommended for the reference population to represent covariate effects. Continuous covariates should be modeled as such, and not "binned" into categories for analysis purposes.

It is recommended that covariates be incorporated into the model in such a way as to preserve positive PK parameters using additive, proportional, exponential, and/or power approaches. The distribution of the covariate should also be determined prior to incorporation into the model.

Covariates should be centered at specific values; i.e., incorporated in such a way as to describe the central tendency of the data. Centering covariate effects may allow for easier interpretation of parameters estimates, and increase the numerical stability of parameter estimation because the null value falls in the middle (presumably) of the covariate distribution instead of at an extreme value. On the rare occasion when the data span a very large range of values (e.g., age in pediatric through elderly subjects), the analyst should seek additional advice, including the clinician, as this input might prove quite valuable to understanding the results of the modeling effort. Finally, it should also be verified that centering the covariate does not allow impossible values to be estimated.

6.4.4 Boundary for Covariate Parameters

In general, setting boundaries on covariate parameters should not be implemented. Instead, the model should be parameterized in such a way as to enforce plausible values for the parameters of interest, such as with the use of a log scale. If boundaries are used instead of a positive scale, a VPC using multivariate normal sampling and other methods may generate negative values when sampling (although a VPC using a bootstrap approach would not be affected). Occasionally, boundaries may need to be set on covariate parameters in order to limit the search space within the NONMEM algorithm to speed convergence and/or avoid physiologically unreasonable estimates.

6.5 Full Covariate and Final Model Interpretation

Once a model which includes all covariates of interest has been developed and minimizes successfully, the analyst must then decipher what information can be obtained from the resulting parameter estimates. Based upon the resulting CI, inferences about the influence of covariate effects on the PK parameters could be made as follows:

- If the CI is sufficiently narrow and includes the null value, the data do not support a covariate effect.
- If the CI is sufficiently narrow and excludes the null, the data appear to support a covariate effect.
- If the CI is very wide but excludes the null value, there appears to be a covariate effect, but the data contain inadequate information to precisely estimate the effect.

The analyst must determine the meaning of "sufficiently narrow" and "very wide" above based upon the clinical relevance and/or importance of a particular parameter. Regardless of which criteria the covariate meets, a statement about the clinical relevance of the effect should be included to aid the interpretation. For example, if a covariate influences CL, the analyst should state the magnitude of that effect. Likewise, a similar statement should be included for covariates that do not support an effect in the analysis. The results could be presented as a Forest plot to show the clinical impact of covariates on key PK or exposure parameters (Cmax, AUC, etc.), such as those examples included in recently approved United States Prescribing Information (USPIs) [34].

6.6 Assessment of Model Adequacy

When covariate model building involves a procedure which generates a full covariate model prior to a final model (e.g., the SCM procedure), it should be verified that the full covariate model represents the greatest reduction in OFV that is possible by inclusion of all the covariates. For that reason, if the OFV of the full covariate model is not lower than the base model, it suggests that the full covariate model is in a local minimum instead of a global minimum.

6.7 Random Effects

Details can be found in 5.6 Random Effects Model Development as the same principles apply to both base and final Pop PK models.

6.8 Additional Topics

6.8.1 Outliers

It is recommended that conditional weighted residuals (CWRES) be implemented to define outliers with |CWRES| > 6. Outliers should be investigated at each stage of model building from the base, full covariate (if implemented), and final models. Outliers should not be excluded as a standard approach without careful inspection of data and models; however, influential outliers could be removed from the dataset and this must be clearly documented in the PMAR. Outliers could be considered influential if a change of greater than 20% in one or more of the key fixed effects parameters is observed when sensitivity analyses are performed; i.e., running models with and without outliers. The results of the sensitivity analysis should be documented in the PMAR.

It is recommended that outliers be retained during the model building process, and then removed only during final model development.

6.8.2 Over-parameterization

During Pop PK model development, analysts are likely to encounter models which fail to minimize successfully. In particular, over-parameterization is often the cause for failure of covariate models to minimize successfully. Minimization termination messages from NONMEM can take several forms, and each requires a different approach for resolution.

- **Inspection of gradients**: The following are pathologic behaviors of gradients which may occur during minimization. Any of these conditions suggest that there is insufficient information in the data for estimation of those parameters, which can be due to either a coding error or lack of data.
    - Any gradients that are 0 at the 0th iteration or the final iteration
    - Gradients that swing from wildly positive to wildly negative; e.g., from 1.0E+07 to -1.0E+07
    - Gradients that do not move much from the initial gradients; e.g., 1.0E+07 to 1.0E+06
- **Initial estimates**: Another possibility for resolving minimization failures is to modify initial estimates to provide the search algorithm with a different starting point. Once the analyst has identified potentially problematic parameters, initial estimates for those parameters may be changed (one at a time) to see if a successful minimization can be obtained. Regardless of which parameters appear to be causing the problems, changing OMEGA initial estimates is often more influential/effective than changing Estimate of Fixed Effect in NONMEM (THETA) initial estimates for resolving problematic runs. Increasing/decreasing OMEGA initial values a few times make the work of changing initial values less cumbersome.
- **Structure of covariate relationships:** If data are truly lacking for a particular covariate relationship, reparameterization of the structure of that relationship is unlikely to resolve the instability. The analyst may try a different structure for those relationships to see if any improvement in stability is appreciated, but minimal effort should be expended to try different structural relationships.
- If the termination message is a boundary condition, the analyst should examine the gradients, find the parameter(s) with zero gradients, and either remove the parameter or relax the boundary conditions.
- If the minimization failure is due to rounding errors, the approach to resolution depends upon the lowest NSIG (number of significant digits) value returned in the NONMEM output. If the lowest NSIG is ≥ 2, the analyst may consider NSIG = 2 on the $EST line. If the lowest NSIG is < 2, the model is likely over-parameterized, and those parameters with low NSIG values should be considered for removal.

- If the minimization error informs the user of an infinite objective function value, there is no general guidance, as this is a case-by-case problem.

6.8.3 Fixing Parameters Based on Prior Information

Setting a parameter to a fixed value in a Pop PK model should be a last resort to increase model stability or parameter estimation. Fixing a parameter to a single value makes a strong statement that the value of this parameter is known precisely, when in fact the opposite may be true. As an alternative to fixing the parameter value, it is recommended that uncertainty about the parameter value be used more formally through use of the PRIOR functionality available in NONMEM 7. PRIOR distributions may be based on knowledge from previous analysis with this compound or other similar compounds or can be subjective, reflecting the analyst's opinion about the range of likely values for the parameter.

It is recommended that, at a minimum, the sensitivity of final inferences be reported for various fixed values of the parameter when reporting the results of models with fixed parameters or when using the PRIOR statement. A rationale for the choice of the fixed value should be provided in the PMAR, and should detail what information was used to inform the choice of the fixed value and whether that value came from a previous analysis of the compound, prior experience with similar compounds, etc. If a subjective prior is used to inform the PRIOR distribution, the analyst should give some rationale and should report the likely range of values for the distribution (analogous to a confidence interval for the prior data).

7. Graphical and Numerical Diagnostics

7.1 Types of Diagnostics

In general, diagnostics can be categorized based on the type of variables being plotted or measured as follows:

- Prediction-based diagnostics (concordance plots); e.g., PRED (Population Predicted value), IPRED (Individual Predicted value),
- Residual-based diagnostics; e.g., WRES (Weighted residuals), CWRES, IWRES,
- Parameter or Empirical Bayes Estimates (EBE)-based diagnostics,
- Simulation-based diagnostics.

7.2 Inferences from Diagnostic Plots

Diagnostic plots are used to guide Pop PK model development. To enable an efficient model development process, it is important to identify the following patterns in these plots [1, 35]:

- lack-of-fit (structural model misspecification),
- violation of the statistical assumptions where the distributions of intra-and inter-individual random effects exhibit:
    - multimodality,
    - asymmetry,
    - non-zero mean, or
    - heterogeneity of variance,
- outliers

It is recommended that the data be appropriately stratified when providing informative diagnostic plots, as follows:

- Consider stratifying diagnostic plots by key design features, such as study and treatment groups (e.g., dose/regimen groups), to avoid masking potential lack-of-fit with respect to these features. Partitioning the plots or using different symbols to denote different studies or treatment groups can be useful to diagnose potential lack-of-fit with respect to these features.
- Consider stratifying diagnostic plots by all included/selected covariates for covariate model building. It is recommended that "before-and-after' plots be provided for all key covariates to show the impact of those covariates.
- Consider stratifying by time if a compound has a long half-life to ensure adequate model performance across various phases of drug disposition.

It is important to note that diagnostic plots will display only non-censored/non-missing data, therefore, the analyst should use caution when interpreting diagnostic plots based on datasets which contain large amounts of censored/missing data.

7.3 Recommended Diagnostics

It is recommended that the following diagnostics be included in the PMAR for key models [3]:

- Assessment of shrinkage of ETA(η) and EPSILON(ε)(7.7.1 Shrinkage of ETA(η) and EPSILON(ε))
- Observations (OBS) versus PRED in both linear and log scale with a line of identity and a regression line (7.5.1 DV (or OBS) versus PRED)
- OBS versus IPRED in both linear and log scale with a line of identity and a regression line (7.5.2 DV (or OBS) versus IPRED)
- WRES or CWRES versus PRED (7.6.2 Weighted Residual versus Predicted Values)
- Absolute Individual weighted residuals (|IWRES|) versus IPRED
- WRES or CWRES versus Time or Time after dose (TIME or TAD) (7.6.1 Weighted Residual versus TIME)
- OBS (or Dependent Variable (DV)), IPRED and PRED concentrations versus Time (overlaid and/or side by side) (7.5.3 Individual Concentration-Time)
- Histogram and/or QQ plot of IWRES and WRES or CWRES (7.6.3 Histogram and/or Quantile-Quantile-Plot (QQ Plot) of Weighted Residuals)
- Visual Predictive Checks (7.8.3 Visual Predictive Checks)

These diagnostics are applied to the base and final models, and could be presented in parallel to demonstrate the improvement from base model to final model after covariate inclusion.

Additional goodness-of-fit plots for the final model could include:

- A scatter plot matrix of EBE of the parameters and/or ETAs (7.7.2 Scatter Matrix Plot of Parameters and ETAs)
- Distributions (e.g., histograms) of EBE of the parameters and/or ETAs
- (7.7.3 Histogram of Parameters and ETAs)
- Plots of EBE of ETAs versus the covariates (7.7.4 ETAs versus Covariates)
- Individual plots that illustrate how well the model describes the data for any given subject

- Additional plots, such as mirror plots, can be presented if justified (7.8.1 Mirror Plots)

7.4 Training and Examples Available For Diagnostic Plots

Xpose is an R-based model building aid for population PK or PK/PD analyses using NONMEM [36]. VPC is implemented within PsN version >2.3.0 and plots of VPC results are available in the R library Xpose version 4.3.2.

7.5 Prediction-Based Diagnostics

7.5.1 DV (or OBS) versus PRED

This plot is used to assess the adequacy of PRED as a measure of the central tendency of the data (DV or OBS). Systematic deviations from concordance suggest that the PRED is not a good measure of central tendency, and further development of the structural and/or statistical features of the model may be warranted.

One of the limitations with this type of graph is that there is no expected pattern. Other than model misspecification, the magnitude of spread around the line of identity will depend on many factors, such as the magnitude of residual and parameter variability, data censoring, etc. Additionally, the regression line does not take into account the heteroscedasticity in the error structure and the fact that the data come from separate individuals. Therefore, these type of plots should be used and evaluated with caution [37]. Simulation approaches to deal with these problems are described in 7.8.1 Mirror Plots.

7.5.2 DV (or OBS) versus IPRED

This plot is used to assess the adequacy of the individual prediction as a measure of central tendency of the individual data (DV or OBS). The magnitude of spread around the line of identity only reflects unexplained residual variability.

Individual data need to be sufficiently informative for the parameters in order for this diagnostic to be informative regarding model misspecification. If individual data are sparse, IPRED will shrink toward the actual observation, resulting in an overfit [16]. The analyst is urged to verify that ε-shrinkage is less than 20% prior to providing DV versus IPRED plots for diagnostic purposes.

A DV versus IPRED plot for the final model which includes covariate effects does not provide diagnostic information regarding the fit with respect to time-invariant covariate effects. This is because the IPRED reflects residual variability, whereas time-invariant (e.g., baseline) covariates attempt to explain inter-individual variability, and hence, IPRED concordance plots show very similar spreads for both the base and final model fits. The Pfizer internal Points for Consideration document recommends that the DV versus IPRED plots be used to evaluate the base model fit only. 'Before-and-after' plots for log(DV) versus log(PRED) should be used for the base model ('before') and the model following addition of covariates ('after') to evaluate the final model fit [1].

7.5.3 Individual Concentration-Time

Individual profiles are very useful in assessing individual fits. DV (OBS), IPRED, and PRED concentration-time profiles should be overlaid or plotted side by side for individual subjects.

7.6 Residual-Based Diagnostics

While concordance plots are often useful for identifying discrepancies in the model fit, residual plots provide a more direct evaluation of these discrepancies (i.e., systematic deviations), with different types of residual plots focusing on either the structural or statistical features of the model. Thus, to assess model adequacy, it is recommended that greater emphasis be placed on using residual plots rather than concordance plots.

7.6.1 Weighted Residual versus TIME

Plots of WRES versus independent variables that influence the structural model are useful for identifying possible structural model misspecification. For a PK model, time is a key independent variable. A horizontal zero reference line and a smooth line should be included within the plot. TIME can be both TAD and continuous time (time in the study). If time covers a long duration, WRES versus log(TIME) should be considered. Generating separate plots for each dose interval when intensive PK samples (e.g., following first and last doses of a multiple dose study) are collected, and using TAD rather than continuous time, can improve resolution.

It is recommended that CWRES be used instead of WRES [38] unless this is not possible. Calculations for WRES are based on the First-Order (FO) approximation method, which is sometimes too crude, and can generate WRES plots that indicate model misspecification when there is none. Even in cases when generation of WRES is implemented with the First-Order Conditional Estimation (FOCE) approximation method, the WRES are still derived using the FO approximation method.

Calculations for CWRES are based on the FOCE approximation method. CWRES is directly related to a term in the FOCE objective function, but can also be calculated using the FO method (using the POST HOC step in NONMEM). Both NONMEM 7 and Xpose provide automated calculation of CWRES for diagnostic plots.

When the FOCE with interaction method (FOCEI) is used, the CWRES versus TIME plot is still informative. Nevertheless, CWRESI (conditional weighted residuals with interaction), which incorporates the interaction terms present in the FOCEI objective function, is now a newly added functionality in NONMEM 7, and should be used to evaluate possible structural model misspecification.

7.6.2 Weighted Residual versus Predicted Values

CWRES or CWRESI versus PRED plots are preferred over WRES versus PRED plots. A horizontal zero reference line and a smooth line should be included within the plot. If a wide range of PRED is generated, the analyst should consider plotting log(PRED) on the x-axis.

|IWRES| versus IPRED plots are used to identify potential residual error model misspecification. A smooth line should be added to the plot. If the constant variance assumption is not violated, the smooth line should be horizontal (zero slope) across the range of IPRED values. The

|IWRES| versus IPRED plot is an effective diagnostic plot to assess the homogeneous (constant) variance assumption.

7.6.3 Histogram and/or Quantile-Quantile Plot of Weighted Residuals

Histograms of the weighted residuals (CWRES and IWRES) are better suited to assess the distributional properties to ensure that the residuals are unimodal and symmetrically distributed about zero. Consideration should be given to generating histograms for both the CWRES and IWRES to assess distributional properties of the intra-individual random effects. Departures from symmetry may be more pronounced in a histogram of the CWRES. In addition, agreement between kernel density smooth line and normal density approximation is desired. Appropriate binning for histograms will depend on the number of observations. Alternatively, IWRES distributions can be visualized and compared to the true normal distribution using quantile-quantile (Q-Q) plots. Note that when ε-shrinkage increases, deviation of the IWRES distribution from normality also increases.

7.7 Empirical Bayes Estimates-Based Diagnostics

EBE-based diagnostics for each parameter rely on investigation of the corresponding ETA (η) for that parameter. The individual parameter estimates are obtained using Bayesian methodology and represent estimates informed by a prior population parameter and residual error distribution, and the individual data.

7.7.1 Shrinkage of ETA(η) and EPSILON(ε)

The phenomenon of EBE variance shrinking towards the population mean is defined as η-shrinkage [16]. Similarly, with informative data, IWRES will approach a normal distribution with zero-mean and unit variance, and as data quantity diminishes, the IWRES distribution will shrink towards zero. This latter phenomenon is defined as ε-shrinkage. When shrinkage is greater than 20%, EBE-based diagnostics are not considered to be informative. As the shrinkage on EBEs impacts not only diagnostics but other relationships; e.g., covariate and exposure-response relationships, the analyst should carefully assess shrinkage and its impact on inferences. When an analysis is performed across studies with varied designs, sometimes only a subset of the data contains substantial information about a particular parameter, and in those cases, it may be useful to stratify the data by study or PK design and calculate the shrinkage for the different stratified groups.

7.7.2 Scatter Matrix Plot of Parameters and ETAs

In order to investigate correlations between parameters, the analyst may consider generating scatter matrix plots. These correlations should be taken into consideration during the covariate model building process.

7.7.3 Histogram of Parameters and ETAs

ETA (η) histograms and Q-Q plots are very informative for assessing the assumption of normality and the appropriateness of the selected parameter variability.

7.7.4 ETAs versus Covariates

These plots are often used for inspection of potential relationships between parameters and covariates, and will vary depending on whether covariates are continuous or categorical. Scatter

plots are often used for continuous variables, whereas box plots are reserved for categorical covariates. Caution is advised when converting continuous variables to categorical variables. Graphical displays should not be used to drive the covariate model building process. The best use of graphical displays is to explore the shape of, rather than to identify, the relationships between covariates and parameters. It is recommended that scatter matrix plots of covariates be provided during the covariate model building process.

7.7.5 Other Parameter Based Diagnostics

A p value from the hypothesis test indicating whether the mean value of EBE (ETABAR) is significantly different from 0 is provided in the NONMEM output. Mean values for the key parameters of interest that are significantly different from zero should be further investigated.

7.8 Simulation-Based Diagnostics

Simulation-based diagnostics enable assessment of model fits via generation of new datasets through simulation, followed by an assessment of the current model versus that simulated data [37, 39, 40]. These methods help assess whether the current model adequately describes the observed data, as well as whether the model is predictive of future data. Simulation-based diagnostics take several forms and address different issues. When performing simulations, the analyst should consider what question is being addressed via the simulation to aid in selection of the simulation method as described below:

- The **VPC** is used to assess whether the current model describes the central tendency and variability in the observed data by comparing the observed median and quantiles to the median and quantiles of the simulated dataset; i.e., data that would arise using the observed data structure if the hypothesized model is true. **Mirror plots** and **Numerical Predictive Check (NPC**) methods are similar simulation-based methodologies. All three of these methods can use the same simulated datasets.
- **Typical Value Predictions** are not used for model diagnostics, but these describe the range of plausible typical values that might be expected given the current parameter estimates and the uncertainty surrounding those estimates.
- **Prediction of future trials** involves simulating new data, possibly using new trial designs and associated new data structures. This provides some idea of the range of plausible trial outcomes given the current model. Critically, prediction involves acknowledging the uncertainty around model parameters, as current model parameters are only estimated from data that has been observed. There are a number of commonly used prediction simulations:
    - **Posterior predictive check (PPC)** is sometimes used for assessing the predictive properties of a model. Typically, the data structure of the observed data is used, but uncertainty in all model parameters is simulated.
    - **Clinical Trial Simulation (CTS)** is used to predict future clinical trial outcomes for a range of clinical trial designs, analytic methods, and decision criteria. In order to be predictive, simulations should reflect the uncertainty in model parameter values.

Table 2 below summarizes the different types of predictive checks and simulations, including how and when to incorporate uncertainty in model parameters.

Table 2. Types of Predictive Checks and Simulation Procedures

Simulation Purpose	Name	Data Structure	THETAs	OMEGAs	SIGMAs
Visual predictive check for model diagnostic [40]	VPC	Data structure from observed data used in modeling	FIXED at MLE (maximum likelihood estimation)	FIXED at MLE	FIXED at MLE
Typical value predictions	EQDM - PTV	One subject per design / covariate combination (if OMEGA, SIGMA=0) or many (100-1000) subjects if averaging over ETAs	Must include uncertainty (Multivariate Normal based on MLE + COV matrix or bootstrap COV matrix)	FIXED at ZERO **or** FIXED at MLE and simulations should average over MANY (100-1000) subjects if ETAs enter the model nonlinearly	FIXED at ZERO
Predictions for future trials	Prediction	Future trial data structure	Must include uncertainty (Multivariate Normal based on MLE + COV matrix or bootstrap COV matrix)	Should include uncertainty (Inverse-Wishart distribution based on bootstrap SEs/COV matrix). Often FIXED at MLE if the model is based on a large number of subjects; i.e., assume a good estimate of OMEGA from the observed data	Should include uncertainty (Inverse-Gamma distribution or Half-Normal based on bootstrap SE(s) / COV matrix). Often fixed at MLE.
Posterior predictive check model diagnostic [39]	PPC	Data structure from observed data used in modeling.	Must include uncertainty (Multivariate Normal based on MLE + COV matrix or bootstrap COV matrix)	Must include uncertainty Inverse-Wishart distribution based on bootstrap SEs / COV matrix	Must include uncertainty Inverse-Gamma distribution or Half-Normal based on bootstrap SE(s) / COV matrix
Clinical trial simulations	CTS	Future trial data structure	Must include uncertainty (Multivariate Normal based on MLE + COV matrix or bootstrap COV matrix)	Should include uncertainty (Inverse-Wishart distribution based on bootstrap SEs/COV matrix). Often FIXED at MLE if the model is based on a large number of subjects; i.e., assume a good estimate of OMEGA from the observed data	Should include uncertainty (Inverse-Gamma distribution or Half-Normal based on bootstrap SE(s) / COV matrix). Often fixed at MLE.

7.8.1 Mirror Plots

Prediction-, residual-, and EBE-based model diagnostics address only deficiencies in the model. Mirror plots are provided with the intention of comparing graphical model diagnostics using the observed data to simulated data, which has been produced using a predefined base/full covariate/final model. In the absence of model misspecification, simulated data should "mirror" the diagnostic plots obtained with the observed data. To define a reference for the real data plot in these types of situations, the model can be used to simulate and analyze new datasets. The same graphical model diagnostics can then be generated (simulated data) and they should look the same as the diagnostics obtained from the observed dataset.

7.8.2 Using Statistics to Assess Model Fit

Standard predictive checks look at whether the model captures features seen in the observed data. It is often useful to check statistics, such as AUC or Cmax, which can be calculated from the simulated data using the model. The same statistics can be calculated from the observed data and, then compared to the simulated statistics. One useful way of comparing the simulated and observed statistics is to graphically display where the observed mean statistic falls within the distribution of means from the simulated data. The analyst can then also report the probability that the true mean (under the assumption that the model is adequate) is greater than (or less than, depending on the question of interest) the observed mean.

7.8.3 Visual Predictive Checks

7.8.3.1 Principles

VPC is a model evaluation method that allows the analyst to assess how well the current model describes the observed data [40, 41]. This is a simulation-based diagnostic which aims to compare the mean and quantiles of the simulated data from the model against the mean and quantiles of the observed data. If the model adequately describes features in the data, then the simulated and observed means and percentiles should correspond closely. It is important to note that the VPC is usually performed by simulations which incorporate the final THETA estimates, and OMEGA and SIGMA values. Generation of new observations from the OMEGA and SIGMA distributions (new ETA and EPS values; i.e., uncertainty in the parameter estimates THETA, OMEGA, and SIGMA) is not included in this process.

The VPC describes the observations that would be expected for a set of individuals using the same set of inputs (dosing records, time, covariates) that were observed in the current data for a given candidate model. It is important to recognize that VPC does not provide a predictive distribution; i.e., it will not describe the range of plausible outcomes for a new, future trial.

7.8.3.2 How to Conduct a Visual Predictive Check

The steps for conducting a VPC are outlined below:

- Using the final parameter estimates for THETA, OMEGA and SIGMA, simulate new observations (DV) for a large number of trials using the observed data structure (dosing records, observation times, covariate values, etc.),
- Calculate the 50th percentile, mean, and other quantiles of interest across all of the simulated data within each bin, (optionally, calculate the median, quantiles within each bin for each simulation, and then calculate the 5th, 95th percentile of those quantiles),

- Compare the mean, median and quantiles of the observed data to the quantiles of the simulated data. This is usually done graphically by plotting the observed data (DV versus TIME) and observed mean / median versus the quantiles of the simulated data for each bin.

7.8.3.3 Points to Consider when Performing a Visual Predictive Check

Several points that need to be considered in order to perform a VPC are described below. Many of these factors are related to each other; e.g., 1) the choice of percentiles for the interval estimates and the number of simulation replicates required to reach a desired level of accuracy, and 2) the number of bins used is related to the number of observations in each bin. The rationale(s) for the number of simulation trials chosen, the percentiles chosen, and how the data will be binned should be stated clearly in the PMAP and PMAR.

- **Number of Simulation Trials**: The number of simulation trials chosen for the VPC depends primarily on the purpose of the exercise and/or the question of interest, i.e., the interest in evaluating the central tendency or the tails of the distribution. For most purposes, 1000 simulation trials should be sufficient. If greater precision is desired, then a higher number of simulations may be required. For example, this may be the case when maximum concentrations drive safety concerns or low concentrations lead to treatment failure.

  The simulation process for VPCs using PsN and NONMEM does not refit the model, so the number of simulations should not be governed by the complexity of the model (assuming the model takes a significant time to estimate parameters). With adequate computing resources (CPU speed and memory size), technical limitations should not drive the choice for the number of simulated trials.

- **Choice of Percentiles**: The choice of which percentiles of observed and simulated data to present depends largely on the amount of observed data per binning interval. Different parts of the model (structural, variability, etc.) are likely to be maximally informed by different percentiles and depend on data richness [42].

  Typically, 90% or 95% percentiles are used, but this choice depends on the number of observations within a bin. If the number of observations is small (10-20 per bin), then the ability of the VPC to show observations outside the percentile range is reduced; e.g., only 1 or 2 points may fall outside of a 90% percentile range (5th – 95th percentile). In this case a lower percentile range, such as 80% (10th – 90th), may be prudent. If the number of observations is large (>100 per bin), then the 90% percentile range should show only 5 observations on either side of the percentile range. When small numbers fall outside of the percentile range, the analyst must take care when interpreting the VPC. It is recommended that 95% percentile range be used only when there are a considerable number of observations per bin (>200).

- **Choice of Binning**: In calculating the quantiles, the analyst may need to "bin" the independent variable (usually TIME) in order to group together values for comparison between simulated and observed data. The advantage of binning is to allow aggregation / summarization without loss of information. For example, if concentration is modeled as a factor of actual time after dosing, the analyst may choose to bin by nominal time post dose, or by sampling window, to combine an observation at 3.9 hours post dose with one at 4.1 hours post dose. However, binning 0-1 hour post dose observations may not adequately describe the changing concentrations in the absorption phase for an orally administered drug. Nominal time points need to fall in the middle of the bin (especially at later sampling points), but the bins should be contiguous.

  The key point for choosing the number and spacing of bins is the ability to group together points which have a similar expected concentration range. Grouping early and late concentrations creates a very wide bin and may dilute the information about the model fit too much. Conversely, having too many bins may decrease the number of points in each bin, making the model fit look potentially worse than it is. In situations for which only sparse data are available; e.g., trough samples, the analyst could consider using other covariates, such as time since first dose or body weight, as the independent variable to create informative VPCs.

- **Stratification:** Stratification can be viewed as another type of binning. It is important to keep in mind that the VPC might not reveal covariate model misspecification if the displays are not stratified on key covariates. Once covariates have been identified as significant (statistically or clinically) in the full covariate/final model, a VPC stratifying by this covariate should be investigated. Stratification by key study design covariates (dose, study, population) should also be investigated to assess model fit for these important aspects. For confirmatory studies, covariates identified a priori should be investigated in the VPC. In order for the VPC to be a robust method for model evaluation, it is important to consider the number of observations in each of the stratification subgroups.

- **Prediction-corrected VPC:** Stratifying across a large number of covariates might hinder the diagnostic value of the VPC. In this case, the analyst may consider using "prediction-(PRED) corrected" VPCs, which normalize the observed and simulated value in each bin by the typical prediction for the median independent variable (e.g., TIME) within each bin [43]. It is recommended that a PRED-corrected VPC be used when covariates are included in the model. It is worth noting that PRED-correction will lose the original scale of the y-axis; i.e., similar to dose-normalization of concentration data.

- **Handling BLQ Values:** The impact of BLQ samples on the interpretation of diagnostics, including the VPC, has been addressed [44].

7.8.3.4 Information Provided by the Visual Predictive Check

Given that the VPC is a simulation-based diagnostic tool that aims to provide graphical comparisons between observed and simulated data, the key is to look for similarities between

observed and simulated data. Below are some points for consideration when assessing whether a model adequately describes the observed data:

- Resemblance of trends / median profiles between observed and simulated data,
- (100-x)% of observations appear outside the percentile range;
  e.g., 5% outside the 2.5^{th} – 97.5^{th} percentile range,
- Observations outside the x% percentile range should be balanced;
  e.g., 2.5% below and 2.5% above the 95% percentile range,
- Observations outside the x% percentile range should be randomly scattered;
  e.g., not aggregated at a particular time point.

In general, if the structural aspects of the model are well-described, then the observed mean / median should be close to the simulated mean / median within each bin. If the variability within the model is well-described, then x% of the data should lie outside the $(100\text{-}x/2)^{th}$ and $x/2^{th}$ percentiles of the simulated data bands.

While the VPC is a useful tool to assess structural model adequacy (both fixed and random effects in mixed effects model), it is less effective for identifying specific sources of model misspecification [41]. Other factors, such as dropout pattern (informative missingness or missing at random) and censored data, should be considered when using the VPC as a diagnostic. A new approach for using the VPC as a diagnostic in the presence of censored and categorical data is provided elsewhere [42, 44].

7.8.3.5 Features of Visual Predictive Check Plots

A list of features which may be considered for inclusion in VPC plots is provided below.

Observations

- Individual scatter points or boxes for each time point
- Lines: smooth, mean / median (default: red solid line)
- Percentile values as lines (default: red broken line)

Simulated data

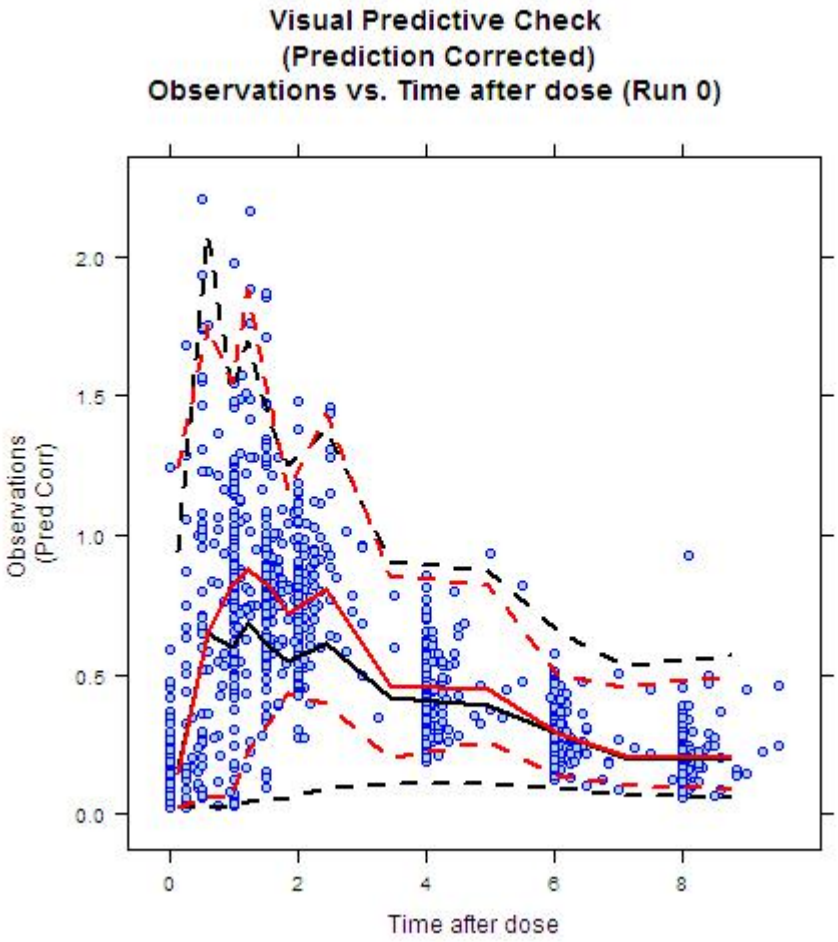
- Line: mean / median (default: black solid line)
- Shaded regions around each percentile range; e.g., 10%, 50% (median), 90%, to illustrate the range of percentile values for each bin across the simulated datasets
- Percentile values as lines (default: black broken line)

7.8.3.6 Recommended Visual Predictive Check Plot Styles

There are several commonly used VPC plot styles, the choice of which depends on the intended audience and the purpose of the plot. Below are two recommended VPC plot styles for non-technical (Figure 1) and more technical (Figure 2) audiences.

Figure 1 could be used to illustrate to a non-technical audience how well the model describes the observed data. When sharing VPC plots outside of the pharmacometrics community, the analyst should carefully consider the ease of communication of the plot contents with the non-technical audience.

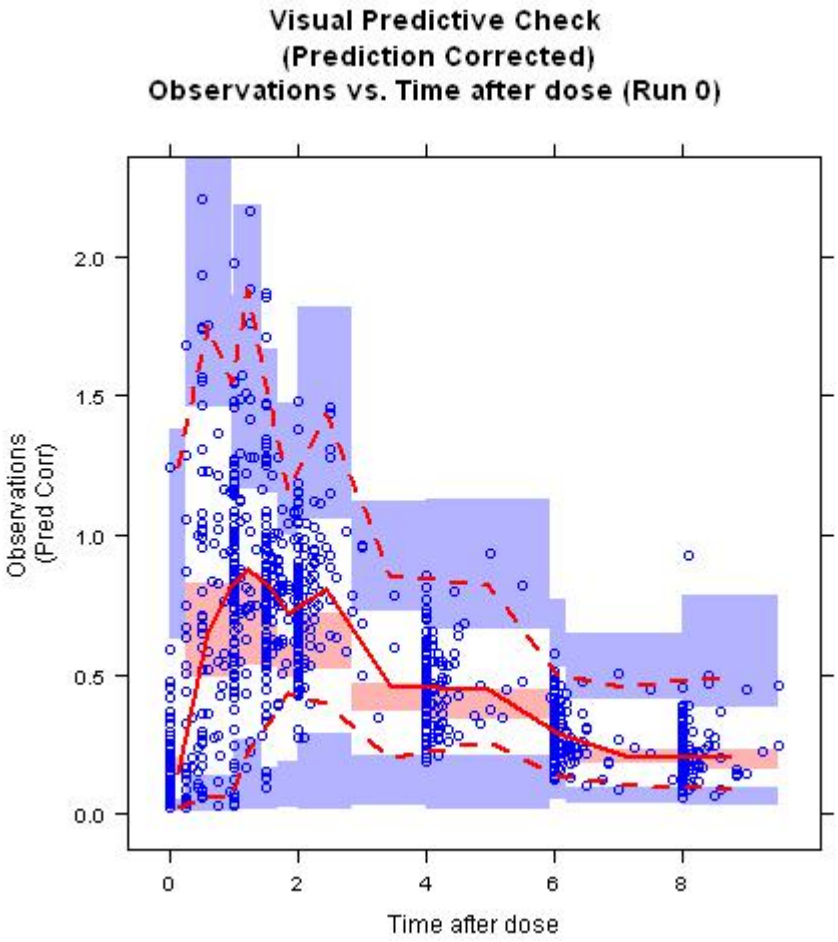
Figure 1. Visual Predictive Check for Non-technical Audiences



Blue circles represent observed data. Red lines represent the percentiles of the observed data. Solid lines represent the median, dashed lines represent the upper and lower percentiles; i.e., 10%, 90%. The black lines represent the median, upper and lower percentiles of the simulated data in each bin (aggregating across simulated trials).

A plot similar to Figure 2 below is preferred for the purpose of discussing model adequacy within the pharmacometrics community.[45] This type of plot shows that the central tendency and quantiles of the observed data are adequately described by the model, and should be the "standard" plot used in the PMAR. However, this visualization style can be complex to communicate outside of the Pharmacometrics community, largely because of the interpretation of the shaded regions (which speak to the range of upper and lower percentile values seen for each bin across the simulated trials). The plot below is the default VPC plot generated within Xpose.

Figure 2. Visual Predictive Check for more Technical Audiences



Blue circles represent the observed data. The lines represent the median, upper and lower percentiles of the observed data. The shaded regions summarise the percentiles within each bin for each simulated trial. For each simulated trial the median, lower and upper percentiles are calculated for each bin, and then a percentile range of these percentiles is shown as the shaded region. If the model adequately describes the data, then all three of the observed (red) lines should fall within the respective shaded regions.

7.8.4 Other Simulation-Based Diagnostics

Simulation-based diagnostics is an evolving area. Additional graphical and numerical methods are being developed to assess model adequacy. Novel graphical methods include the Standardized VPC [46]. Numerical predictive checks, including the normalized prediction distribution error (NPDE) are less frequently used, but are an emerging area [47]. Numerical predictive checks characterize model appropriateness by means of prediction intervals which are derived via simulation from final model parameters and parameter distribution estimates. Based on simulated data, prediction intervals are created for each unique observation. The simulated observations for each particular given observation are ranked and prediction intervals are created. In theory, if the model is correct, the probability of finding the actual observation outside the percentile range (e.g., 90% percentile range) is the same as a 5% chance that that observation will be above or below the prediction interval. NPDE presents an improved version

of prediction discrepancies, as NPDE takes into account repeated observations within a subject. These diagnostics are available using standard tools (PsN).

8. Conclusions

The importance and necessity of implementing a systematic, streamlined, and standardized approach to optimize and harmonize the processes which contribute to the Pop PK analysis cannot be overstated. As this is an area of continually evolving science and technology, this guidance is to be considered a "living document" that will be updated and revised periodically. The Pfizerpedia "PK/PD for beginners" contains information about the Pop PK Guidance, in addition to the workflow and tools needed to conduct Pop PK analyses. The Wiki page aims to facilitate live discussions within the Pfizer Global Clinical Pharmacology community. Discussion topics within the Wiki page will be reviewed periodically, and relevant content will be included in future versions and updates to this guidance.

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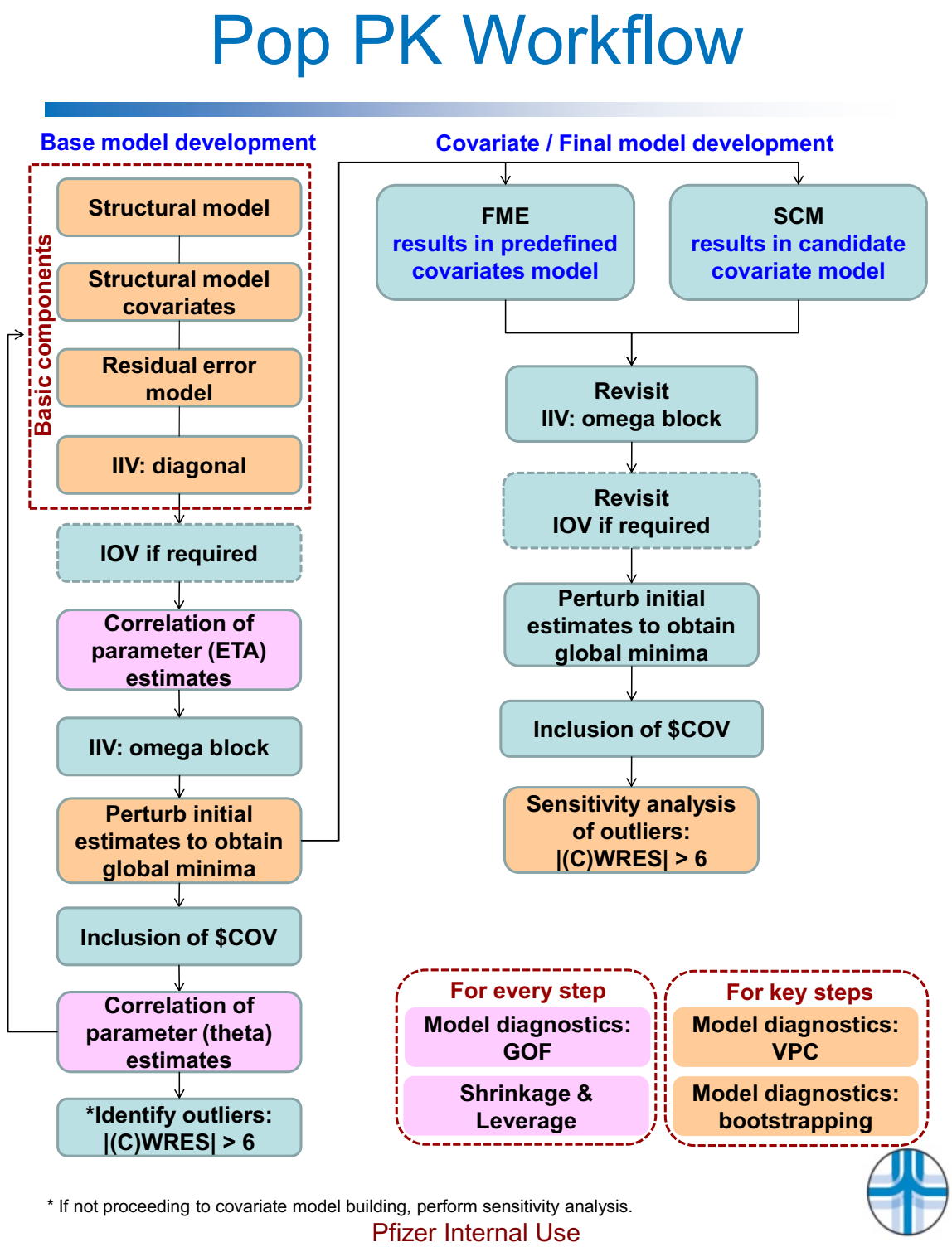
Appendix A. Population Pharmacokinetic Analysis Workflow